

Q2. What is a practical way to insert images into Markdown?

Question

How should I insert screenshots or figures into a .md file and organize their image files?

For this repository, save images in docs/images/ and reference them with relative paths. This keeps a chapter and its figures portable when the repository is copied or cloned.

1. Save the image, then reference it

The repository already contains this example:

```
research-coding-handbook/  
├── README.md  
└── docs/  
    ├── 01-spyder-function-inspection.md  
    ├── 02-markdown-images.md  
    └── images/  
        └── 01-generated-data.png
```

In a chapter inside docs/, write:

```
![Local and global random samples compared with an exponential curve.](images/01-generated-data.png)
```

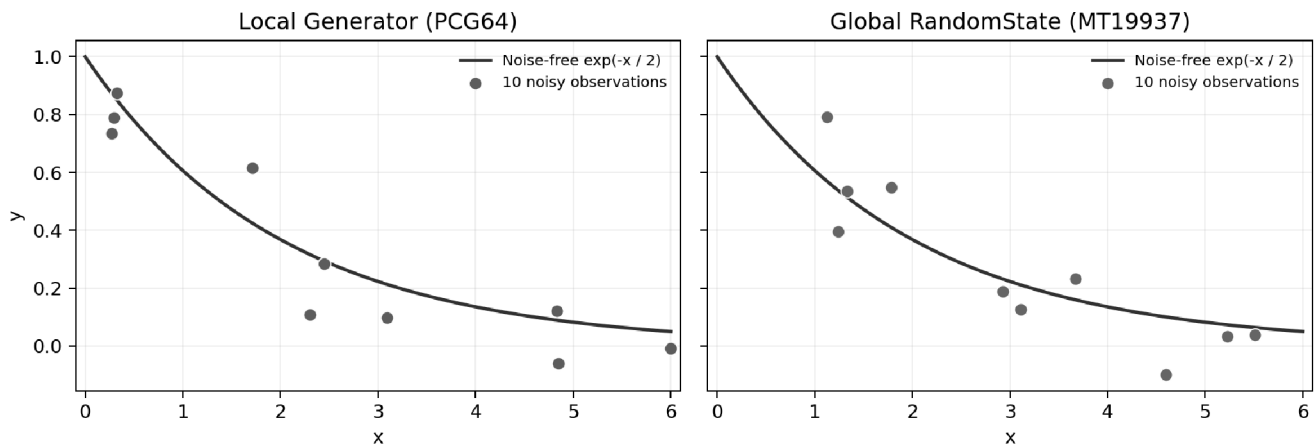
In the root README.md, write:

```
![Local and global random samples compared with an exponential curve.](docs/images/01-generated-data.png)
```

The path starts from the directory containing the Markdown file. In these examples, images/ means a child directory and ../ would mean one directory up. Use forward slashes / in links, including on Windows. The text in brackets is **alternative text**, a short description of the image. See [GitHub: images and relative links](#)^[1].

2. See an actual embedded figure

Same seed = 5; different random streams



Noise standard deviation = 0.12; arrays x and y have shape (10, 1)

Figure: Both versions use seed 5, ten observations, and noise standard deviation 0.12.

The image above is a real file generated from the [Q1 example](#). To regenerate it from the repository root, use an environment with NumPy and Matplotlib:

```
python examples/02_plot_generated_data.py
```

The script writes docs/images/01-generated-data.png; running it again updates that figure.

3. Insert a Spyder screenshot in VS Code

1. Capture the relevant Spyder panes and save the screenshot as a PNG in docs/images/, for example 01-spyder-local-breakpoint.png.
2. Open the intended .md file. Run **Markdown: Insert Image from Workspace** from the command palette and select the saved image. Alternatively, drag it over the Markdown editor, then hold **Shift** as you drop it.
3. Add descriptive alternative text and a short caption stating where execution paused and what the image demonstrates.
4. On Windows, press **Ctrl+Shift+V** to preview the Markdown, or press **Ctrl+K**, then **V**, for a preview beside the editor.

Pasting image data is also supported. Check where the editor saved the image; VS Code's `markdown.copyFiles.destination` setting controls that destination. See [VS Code: Markdown images and preview](#) ^[2].

For Q1, a useful screenshot shows the paused `return x, y` line and the Variable Explorer entries for `x`, `y_true`, `noise`, and `y` together.

4. Control display width when needed

For a renderer that supports raw HTML, you can use:

```

```

Here the path is for a Markdown file inside `docs/`; a root README needs `docs/images/01-generated-data.png`. HTML rendering varies by viewer, so check the target preview. Ordinary Markdown image syntax is the more portable starting point.

5. File and Git conventions

- Use descriptive filenames, such as `01-spyder-local-breakpoint.png`. PNG is a practical choice for screenshots; for generated plots, retain the script so labels or resolution can be changed later.
- Keep the image and its Markdown reference together in version control when the image should be shared. Ignoring a Markdown file does not automatically ignore the images it references.
- The two language versions can reference the same image; each supplies its own alternative text and caption. This repository ignores `README.zh-CN.md` and `docs/*.zh-CN.md`, while the shared example PNG remains eligible for Git tracking.
- If an image does not display, check that it exists, that the path is relative to the current `.md` file, and that filename capitalization matches. Also check that you wrote `![description](path)`: omitting `!` produces a link instead of an image.

References

1. <https://docs.github.com/en/get-started/writing-on-github/getting-started-with-writing-and-formatting-on-github/basic-writing-and-formatting-syntax#images>
2. <https://code.visualstudio.com/docs/languages/markdown>